



Researcher
in the loop

**Propose and
revise ideas**
between campaigns

Generalize and prove
from discovered codes



Machine-check proofs

task specification

inspect codes

01 TARGET

n	$d_\ell - d_u$
7	5 - 6
9	7 - 8
11	9 - 10

Reference tables
Known bounds

02 EVOLVE

LLM evolves
a program



Exact evaluator

feedback

03 VERIFY



SAT + DRAT
proof check

Certify nonexistence

04 DISCOVER



Tighten code bounds

$[[11, 1, 10; 8]]_2$

115

codes

10

exclusions

nonexistence targets